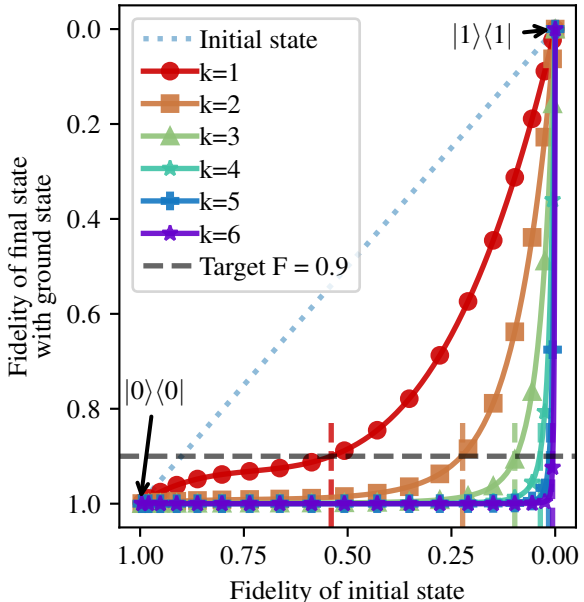
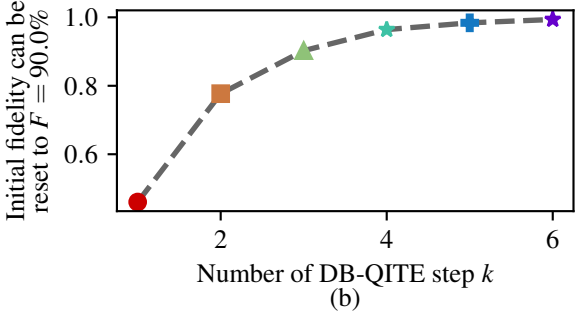
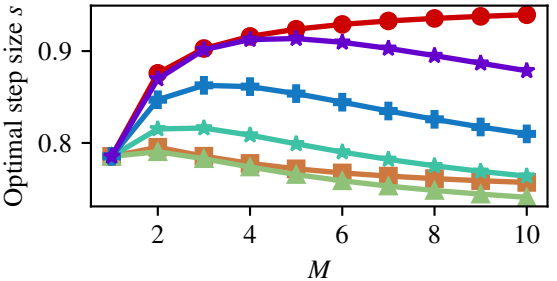

Initial state
k=1
k=2
k=3
k=4
k=5
k=6
Target F = 0.9
|1⟩⟨1|
|0⟩⟨0|
Fidelity of final state with ground state
Fidelity of initial state
0.0
0.2
0.4
0.6
0.8
1.0
1.00
0.75
0.50
0.25
0.00


Initial fidelity can be reset to $F = 90.0\%$
Number of DB-QITE step k
1.0
0.8
0.6
2
4
6

(b)


Optimal step size s
M
0.9
0.8
2
4
6
8
10